

Chapter 8 Test model

Multiple Choice Questions (70 Questions)

1. Which of the following is **not** a unary relational operation in relational algebra?
 - a. SELECT
 - b. PROJECT
 - c. JOIN
 - d. RENAME

2. The symbol used to denote the SELECT operation in relational algebra is:
 - a. π
 - b. ρ
 - c. σ
 - d. $\times$

3. Which property ensures that the result of any relational algebra operation is also a relation?
 - a. Associativity
 - b. Closure
 - c. Commutativity
 - d. Distributivity

4. What is the result of applying the PROJECT operation on a relation that includes a key attribute?
 - a. Fewer tuples than the original
 - b. Same number of tuples as the original
 - c. Duplicate tuples are retained
 - d. The schema is altered unpredictably

5. In relational algebra, the RENAME operation is denoted by:
 - a. σ
 - b. π

- c. ρ
 - d. U
-

6. Which of the following set operations **requires** the operand relations to be type compatible?
- a. CARTESIAN PRODUCT
 - b. JOIN
 - c. UNION
 - d. DIVISION
-

7. The CARTESIAN PRODUCT of two relations R (with 5 tuples) and S (with 3 tuples) results in:
- a. 8 tuples
 - b. 15 tuples
 - c. 2 tuples
 - d. 0 tuples
-

8. Which operation is used to combine related tuples from two relations based on a condition?
- a. UNION
 - b. CARTESIAN PRODUCT
 - c. JOIN
 - d. INTERSECTION
-

9. An EQUIJOIN is a JOIN where the join condition contains only:
- a. Inequality comparisons
 - b. Logical ORs
 - c. Equality comparisons
 - d. Aggregate functions
-

10. The NATURAL JOIN operation automatically joins relations based on:
- a. All attributes with the same name
 - b. Primary keys only
 - c. Foreign keys only
 - d. User-specified theta conditions

11. Which of the following is **not** part of the complete set of relational operations?

- a. SELECT
- b. PROJECT
- c. JOIN
- d. RENAME

12. The DIVISION operation $R(Z) \div S(X)$ yields a relation over which set of attributes?

- a. X
- b. Z
- c. $Z - X$
- d. $X \cup Z$

13. In the expression $\sigma_{DNO=5}(EMPLOYEE)$, what is the selection condition?

- a. EMPLOYEE
- b. DNO
- c. $DNO = 5$
- d. σ

14. Which of the following is true about the PROJECT operation?

- a. It preserves all duplicate tuples
- b. It can increase the number of tuples
- c. It always removes duplicates
- d. It changes the relation name automatically

15. The expression $\pi_{FNAME, LNAME}(\sigma_{SALARY > 50000}(EMPLOYEE))$ is an example of:

- a. A query tree
- b. A single relational algebra expression
- c. An outer union
- d. A recursive closure

16. Which operation is necessary before performing a NATURAL JOIN if the joining attributes have different names?

- a. SELECT

- b. PROJECT
 - c. RENAME
 - d. UNION
-

17. What is the main disadvantage of using CARTESIAN PRODUCT alone?
- a. It eliminates all tuples
 - b. It produces meaningless combinations
 - c. It requires type compatibility
 - d. It cannot be used with SELECT
-

18. Which of the following is **not** a binary relational operation?
- a. JOIN
 - b. DIVISION
 - c. SELECT
 - d. SET DIFFERENCE
-

19. In relational algebra, the attribute names in the result of $R \cup S$ are taken from:
- a. S
 - b. The intersection of R and S
 - c. R
 - d. A new schema
-

20. Which of the following operations is **not commutative**?
- a. UNION
 - b. INTERSECTION
 - c. SET DIFFERENCE
 - d. Both a and b
-

21. The expression $R \bowtie_{\theta} S$ where θ is a general Boolean condition is called:
- a. EQUIJOIN
 - b. NATURAL JOIN
 - c. Theta-join
 - d. Outer join
-

22. Which aggregate function would you use to count the number of employees in the EMPLOYEE relation?
- a. SUM
 - b. AVG
 - c. COUNT
 - d. MAX
-

23. In the notation $\text{DNO } \mathcal{F}_{\text{COUNT SSN, AVERAGE SALARY}}(\text{EMPLOYEE})$, what does DNO represent?
- a. Aggregation function
 - b. Grouping attribute
 - c. Result relation name
 - d. Selection condition
-

24. Which of the following cannot be expressed using basic relational algebra?
- a. Selection with compound conditions
 - b. Recursive closure
 - c. Set difference
 - d. Projection
-

25. The LEFT OUTER JOIN ensures that:
- a. All tuples from the right relation appear
 - b. Only matching tuples appear
 - c. All tuples from the left relation appear
 - d. Nulls are never used
-

26. The OUTER UNION operation is used when:
- a. Relations are type compatible
 - b. Relations are not type compatible but partially compatible
 - c. Performing recursive queries
 - d. Aggregating numeric data
-

27. In the COMPANY database, to find employees who have **no** dependents, which operation is essential?
- a. UNION

- b. INTERSECTION
 - c. SET DIFFERENCE
 - d. CARTESIAN PRODUCT
-

28. The query tree is primarily used for:

- a. Data entry
 - b. Query optimization
 - c. Schema design
 - d. User authentication
-

29. Which of the following expressions correctly retrieves the names of employees working in department 5?

- a. $\sigma_{DNO=5}(\pi_{FNAME, LNAME}(EMPLOYEE))$
 - b. $\pi_{FNAME, LNAME}(\sigma_{DNO=5}(EMPLOYEE))$
 - c. $\rho_{RESULT}(DNO=5, EMPLOYEE)$
 - d. $\mathcal{F}_{DNO=5}(EMPLOYEE)$
-

30. The symbol $\mathcal{F}$ is used in relational algebra to denote:

- a. Selection
 - b. Projection
 - c. Aggregation
 - d. Division
-

31. Which of the following is **true** about the SELECT operation?

- a. It can increase the number of tuples
 - b. It changes the schema
 - c. It is not commutative
 - d. It is idempotent and commutative
-

32. If R has schema (A,B,C) and S has schema (C,D,E), then $R \bowtie S$ (natural join) results in a relation with schema:

- a. (A,B,C,D,E)
- b. (A,B,C,C,D,E)

- c. (A,B,D,E)
 - d. (C)
-

33. The expression $(R - S) \cup (S - R)$ represents:

- a. $R \cap S$
 - b. Symmetric difference of R and S
 - c. $R \cup S$
 - d. $R \div S$
-

34. Which operation would you use to find employees who work in **both** department 4 and department 5 (assuming an employee can work in multiple departments via a separate relation)?

- a. UNION
 - b. INTERSECTION
 - c. DIFFERENCE
 - d. DIVISION
-

35. In relational algebra, duplicate elimination is enforced in:

- a. SELECT only
 - b. PROJECT only
 - c. UNION, INTERSECTION, DIFFERENCE, and PROJECT
 - d. CARTESIAN PRODUCT only
-

36. Which of the following is a **very hard** distinction: The result of $R \div S$ includes a tuple t if:

- a. t appears in R and not in S
 - b. t appears in R with every tuple of S
 - c. t is the key of R
 - d. t matches at least one tuple in S
-

37. Consider two relations: $R(A,B)$ and $S(B,C)$. Which expression is equivalent to $R \bowtie S$?

- a. $\sigma_{R.B = S.B}(R \times S)$
- b. $\pi_{A,B,C}(R \cup S)$
- c. $R - S$
- d. $\mathcal{F}_B(R \times S)$

38. To rename only the attributes of a relation $R(A,B,C)$ to (X,Y,Z) , you would use:

- a. $\rho_S(R)$
 - b. $\rho_{(X,Y,Z)}(R)$
 - c. $\rho_{S(X,Y,Z)}(R)$
 - d. $\pi_{X,Y,Z}(R)$
-

39. Which of the following is **not** an aggregate function in extended relational algebra?

- a. COUNT
 - b. TOTAL
 - c. AVG
 - d. MIN
-

40. Recursive closure is needed to retrieve:

- a. Direct supervisees only
 - b. All supervisees at all levels
 - c. Department managers
 - d. Employee dependents
-

41. The FULL OUTER JOIN retains:

- a. Only matching tuples
 - b. All tuples from left relation
 - c. All tuples from right relation
 - d. All tuples from both relations, padding with nulls as needed
-

42. In the expression $\rho_{\text{RESULT}(F,L,S)}(\pi_{\text{FNAME,LNAME,SALARY}}(\text{EMPLOYEE}))$, what is being renamed?

- a. Relation name only
 - b. Attribute names only
 - c. Both relation and attribute names
 - d. Nothing; this is invalid syntax
-

43. Which of the following is **false**?

- a. PROJECT is not commutative

- b. SELECT is commutative
 - c. UNION is associative
 - d. CARTESIAN PRODUCT requires type compatibility
-

44. The number of tuples after a PROJECT operation is always:

- a. Greater than or equal to the original
 - b. Less than or equal to the original
 - c. Exactly the same
 - d. Zero
-

45. Which operation is used to simulate INTERSECTION using other basic operations?

- a. $R \cap S = (R \cup S) - ((R - S) \cup (S - R))$
 - b. $R \cap S = R - (R - S)$
 - c. $R \cap S = \sigma_{\text{true}}(R \times S)$
 - d. Both a and b
-

46. In the COMPANY schema, to get the manager's name for each department, you must join:

- a. DEPARTMENT and DEPENDENT
 - b. EMPLOYEE and DEPENDENT
 - c. DEPARTMENT and EMPLOYEE on MGRSSN = SSN
 - d. PROJECT and EMPLOYEE
-

47. The expression $\pi_{\text{SSN}}(\text{EMPLOYEE}) - \pi_{\text{ESSN}}(\text{DEPENDENT})$ finds:

- a. Employees with dependents
 - b. Dependents without employees
 - c. Employees without dependents
 - d. All SSNs
-

48. Which of the following is a **very hard** question: The DIVISION operation is the inverse of:

- a. UNION
- b. CARTESIAN PRODUCT
- c. JOIN
- d. PROJECTION

49. In query optimization, a query tree's leaf nodes represent:

- a. Intermediate results
 - b. Algebraic operations
 - c. Base relations
 - d. Final output
-

50. Which symbol denotes the NATURAL JOIN in the textbook?

- a. $\bowtie$
 - b. $\bowtie_{\theta}$
 - c. $*$
 - d. $\times$
-

51. The condition for type compatibility **does not** require:

- a. Same number of attributes
 - b. Same attribute names
 - c. Compatible domains
 - d. All of the above are required
-

52. Which operation would be used to combine STUDENT(Name, ID) and INSTRUCTOR(Name, ID, Rank) into a single relation with nulls for non-matching roles?

- a. EQUIJOIN
 - b. NATURAL JOIN
 - c. OUTER UNION
 - d. THETA JOIN
-

53. The expression $\mathcal{F}_{\text{COUNT SSN, AVERAGE Salary}}(\text{EMPLOYEE})$ returns:

- a. One tuple with two values
 - b. Multiple tuples grouped by department
 - c. An error—grouping attribute missing
 - d. All employee salaries
-

54. Which of the following cannot be expressed without grouping?

- a. Total salary of all employees

- b. Average salary per department
 - c. Highest salary in company
 - d. Number of employees
-

55. In relational algebra, relations are treated as:

- a. Lists
 - b. Bags
 - c. Sets
 - d. Trees
-

56. The assignment operator in relational algebra sequences is denoted by:

- a. =
 - b. :=
 - c. $\leftarrow$
 - d. $\rightarrow$
-

57. Which operation is **not** part of the original relational algebra but added later?

- a. SELECT
 - b. PROJECT
 - c. AGGREGATE FUNCTIONS
 - d. UNION
-

58. To list female employees and their dependents, the critical step after CARTESIAN PRODUCT is:

- a. PROJECT
 - b. RENAME
 - c. SELECT with $SSN = ESSN$
 - d. UNION
-

59. The result of $R \div S$ is empty if:

- a. S is empty
- b. R is empty
- c. There exists a tuple in S with no matching combination in R
- d. All of the above

-
60. Which of the following is **very hard**: In the division $R(Z) \div S(X)$, if S is empty, the result is:
- a. Empty
 - b. All possible Y values (where $Y = Z - X$)
 - c. Undefined
 - d. R itself
-

61. The expression $\rho_S(R)$ changes:
- a. Only attribute names
 - b. Only relation name
 - c. Both
 - d. Neither
-

62. Which of the following is equivalent to $\sigma_{\text{cond1} \wedge \text{cond2}}(R)$?
- a. $\sigma_{\text{cond1}}(\sigma_{\text{cond2}}(R))$
 - b. $\sigma_{\text{cond2}}(\sigma_{\text{cond1}}(R))$
 - c. Both a and b
 - d. Neither
-

63. In the COMPANY database, the attribute that links EMPLOYEE to DEPENDENT is:
- a. SSN and ESSN
 - b. DNO and DNUMBER
 - c. SUPERSSN and SSN
 - d. MGRSSN and SSN
-

64. Which operation is used to remove unwanted columns?
- a. SELECT
 - b. PROJECT
 - c. RENAME
 - d. UNION
-

65. The JOIN operation is a combination of:
- a. SELECT and UNION

- b. CARTESIAN PRODUCT and SELECT
 - c. PROJECT and RENAME
 - d. DIVISION and DIFFERENCE
-

66. Which of the following is **not** a property of the PROJECT operation?
- a. Eliminates duplicates
 - b. Vertical partitioning
 - c. Can reduce the number of tuples
 - d. Always commutative
-

67. In a THETA JOIN, the condition can include:
- a. Only =
 - b. Only <, >
 - c. Any Boolean expression
 - d. Only AND of equalities
-

68. The OUTER UNION of R(A,B) and S(A,C) results in a relation with schema:
- a. (A,B,C)
 - b. (A,B)
 - c. (A,C)
 - d. (A,B,A,C)
-

69. Which of the following queries **requires** the use of RENAME to avoid ambiguity?
- a. Self-join on EMPLOYEE to find supervisors
 - b. PROJECT on EMPLOYEE
 - c. SELECT on DEPARTMENT
 - d. UNION of two projections
-

70. Which of the following is **very hard**: The minimal complete set of relational algebra operations consists of:
- a. SELECT, PROJECT, UNION, INTERSECTION, CARTESIAN PRODUCT
 - b. SELECT, PROJECT, UNION, DIFFERENCE, RENAME
 - c. SELECT, PROJECT, UNION, DIFFERENCE, CARTESIAN PRODUCT
 - d. SELECT, PROJECT, JOIN, DIVISION, AGGREGATE

Answer Key – MCQs (1–70)

1. c
2. c
3. b
4. b
5. c
6. c
7. b
8. c
9. c
10. a
11. c
12. c
13. c
14. c
15. b
16. c
17. b
18. c
19. c
20. c
21. c
22. c
23. b
24. b
25. c
26. b
27. c
28. b
29. b
30. c
31. d
32. a
33. b
34. b
35. c
36. b

- 37. a
- 38. b
- 39. b
- 40. b
- 41. d
- 42. b
- 43. d
- 44. b
- 45. d
- 46. c
- 47. c
- 48. b
- 49. c
- 50. c
- 51. b
- 52. c
- 53. a
- 54. b
- 55. c
- 56. c
- 57. c
- 58. c
- 59. d
- 60. b
- 61. b
- 62. c
- 63. a
- 64. b
- 65. b
- 66. d
- 67. c
- 68. a
- 69. a
- 70. c

True/False Questions (30 Questions)

71. The SELECT operation can increase the number of tuples in a relation.

- a. True
 - b. False
-

72. The PROJECT operation always preserves the number of tuples.

- a. True
 - b. False
-

73. Relational algebra is closed under its operations.

- a. True
 - b. False
-

74. The CARTESIAN PRODUCT requires the operand relations to be type compatible.

- a. True
 - b. False
-

75. UNION, INTERSECTION, and SET DIFFERENCE all require type compatibility.

- a. True
 - b. False
-

76. The NATURAL JOIN automatically removes one copy of each join attribute.

- a. True
 - b. False
-

77. The DIVISION operation is commutative.

- a. True
 - b. False
-

78. RENAME is only used to change relation names, not attribute names.

- a. True
- b. False

79. A THETA JOIN is a generalization of the EQUIJOIN.

- a. True
 - b. False
-

80. Aggregate functions are part of the basic relational algebra.

- a. True
 - b. False
-

81. The LEFT OUTER JOIN keeps all tuples from the right relation.

- a. True
 - b. False
-

82. In relational algebra, relations may contain duplicate tuples.

- a. True
 - b. False
-

83. The expression $\sigma_{\text{cond1}}(\sigma_{\text{cond2}}(R))$ is equivalent to $\sigma_{\text{cond2 AND cond1}}(R)$.

- a. True
 - b. False
-

84. PROJECT is commutative.

- a. True
 - b. False
-

85. The result of $R - S$ is the same as $S - R$.

- a. True
 - b. False
-

86. Recursive closure can be expressed in basic relational algebra.

- a. True

b. False

87. The OUTER UNION can handle relations that are not fully type compatible.

- a. True
 - b. False
-

88. In the COMPANY database, MGRSSN in DEPARTMENT references SSN in EMPLOYEE.

- a. True
 - b. False
-

89. The symbol π is used for the SELECT operation.

- a. True
 - b. False
-

90. The complete set of relational operations includes JOIN as a primitive.

- a. True
 - b. False
-

91. Grouping is required to compute aggregate functions per group.

- a. True
 - b. False
-

92. The query tree is used during logical query optimization.

- a. True
 - b. False
-

93. INTERSECTION can be expressed using UNION and DIFFERENCE.

- a. True
 - b. False
-

94. In DIVISION, if S has a tuple not matched in R, that eliminates candidate results.

- a. True
 - b. False
-

95. RENAME is never necessary in relational algebra expressions.

- a. True
 - b. False
-

96. The expression $R \div S$ is defined only when the attributes of S are a subset of R's attributes.

- a. True
 - b. False
-

97. FULL OUTER JOIN is the same as EQUIJOIN.

- a. True
 - b. False
-

98. COUNT in relational algebra removes duplicates before counting.

- a. True
 - b. False
-

99. The assignment operator ($\leftarrow$) is part of relational algebra syntax for sequences.

- a. True
 - b. False
-

100. A NATURAL JOIN on relations with no common attributes results in a CARTESIAN PRODUCT.

- a. True
 - b. False
-

Answer Key – True/False (71–100)

- 71. b
 - 72. b
 - 73. a
 - 74. b
 - 75. a
 - 76. a
 - 77. b
 - 78. b
 - 79. a
 - 80. b
 - 81. b
 - 82. b
 - 83. a
 - 84. b
 - 85. b
 - 86. b
 - 87. a
 - 88. a
 - 89. b
 - 90. b
 - 91. a
 - 92. a
 - 93. a
 - 94. a
 - 95. b
 - 96. a
 - 97. b
 - 98. b
 - 99. a
 - 100. a
-